

FPGA100T / MINIMAL CORE

CONNECTED CORE SCHEMATIC / PCB ROUTING IN PROGRESS / DRAFT

Goal: power the FPGA, program it through the shared cable, and boot from flash.

HOW THE CORE WORKS



J1, J2, J3, all test points, reset button and LED removed.

Arrows above explain functions; electrical connections are on the linked pages.

WHAT IS VERIFIED

419 assigned physical pin endpoints match the exported schematic netlist.

This does not prove power quality, routability, timing closure or hardware operation.

SCHEMATIC PAGES

02	FPGA banks 14 and 15
03	FPGA banks 16 and 34
04	FPGA bank 35
05	FPGA configuration and analog
06	FPGA power and ground
07	Cable input capacitors / 12 V draft
08	Core supply / about 1.0 V
09	Auxiliary supply / 1.8 V
10	Configuration supply / 3.3 V
11	ASIC bank supply / 1.5 V
12	Core and BRAM capacitors
13	Auxiliary and ADC capacitors
14	3.3 V bank capacitors
15	1.5 V bank capacitors
16	Flash memory
17	JTAG and fixed mode straps
18	Reference clock
19	Electrical-rule power markers
20	J4 / shared power and JTAG connector

117 APPLICATION SIGNALS

Count confirmed; balls and connectors unassigned.

XEM8310 receiver; cable data protocol remains open.

ENGINEERING DRAFT – NOT FOR FABRICATION

Sheet: /
File: FPGA100T_Minimal.kicad_sch

Title: FPGA100T core / power, boot, clock and JTAG

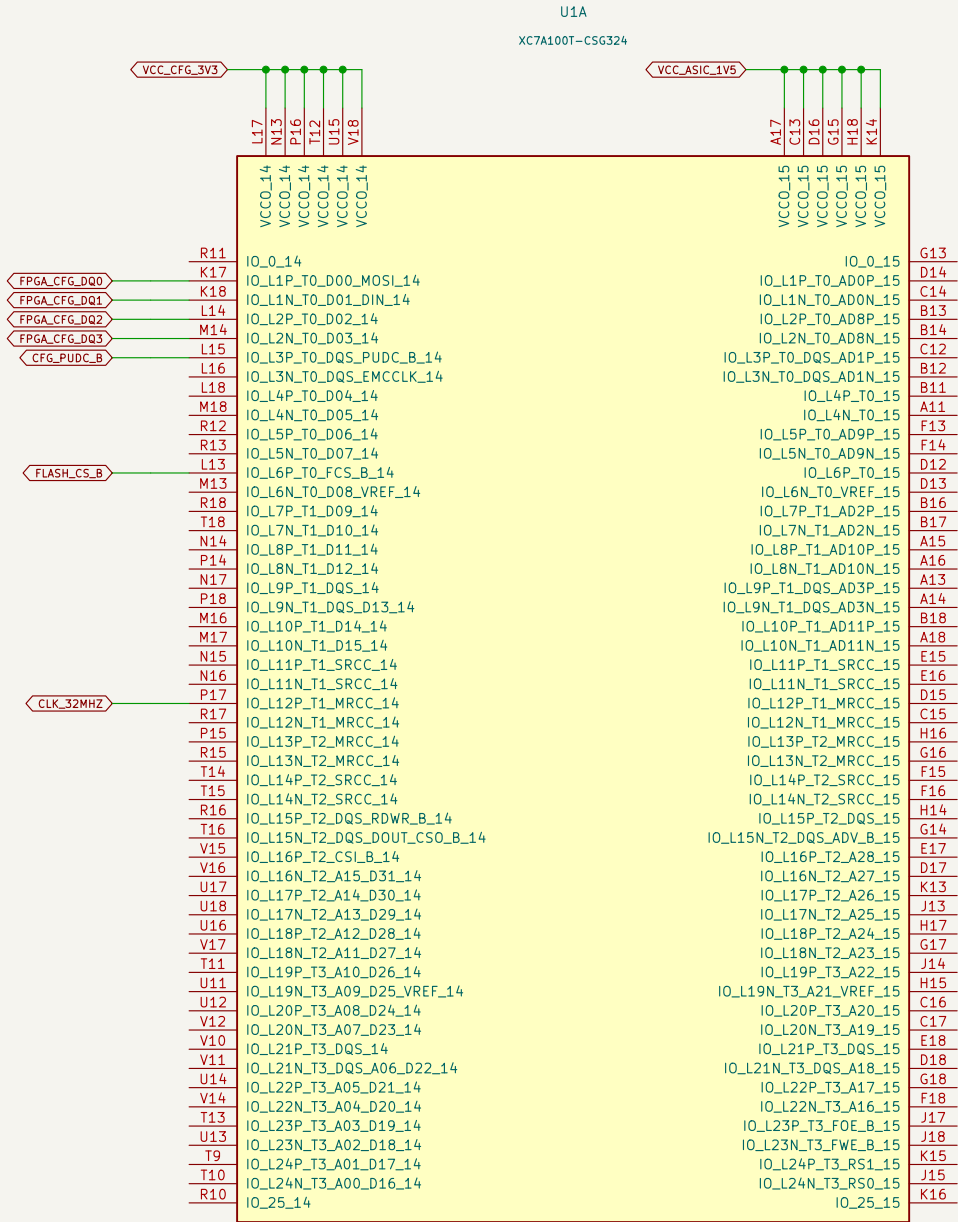
Size: A3 Date: 2026-09-26

Rev: MINIMAL 0.3

SIZE: A3	
KiCad E.D.A. 10.0.6	

U1 / XC7A100T-CSG324 / I/O banks 14 and 15

15 x 15 mm CSG324. Unassigned I/O pins are deliberately open. The 117-signal ASIC interface is a confirmed budget; physical ASIC and receiver assignments remain deferred.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /U1 banks 14 and 15/

File: fpga_unit_01.kicad_sch

Title: U1 / XC7A100T-CSG324 / I/O banks 14 and 15

Size: A3

Date: 2026-09-26

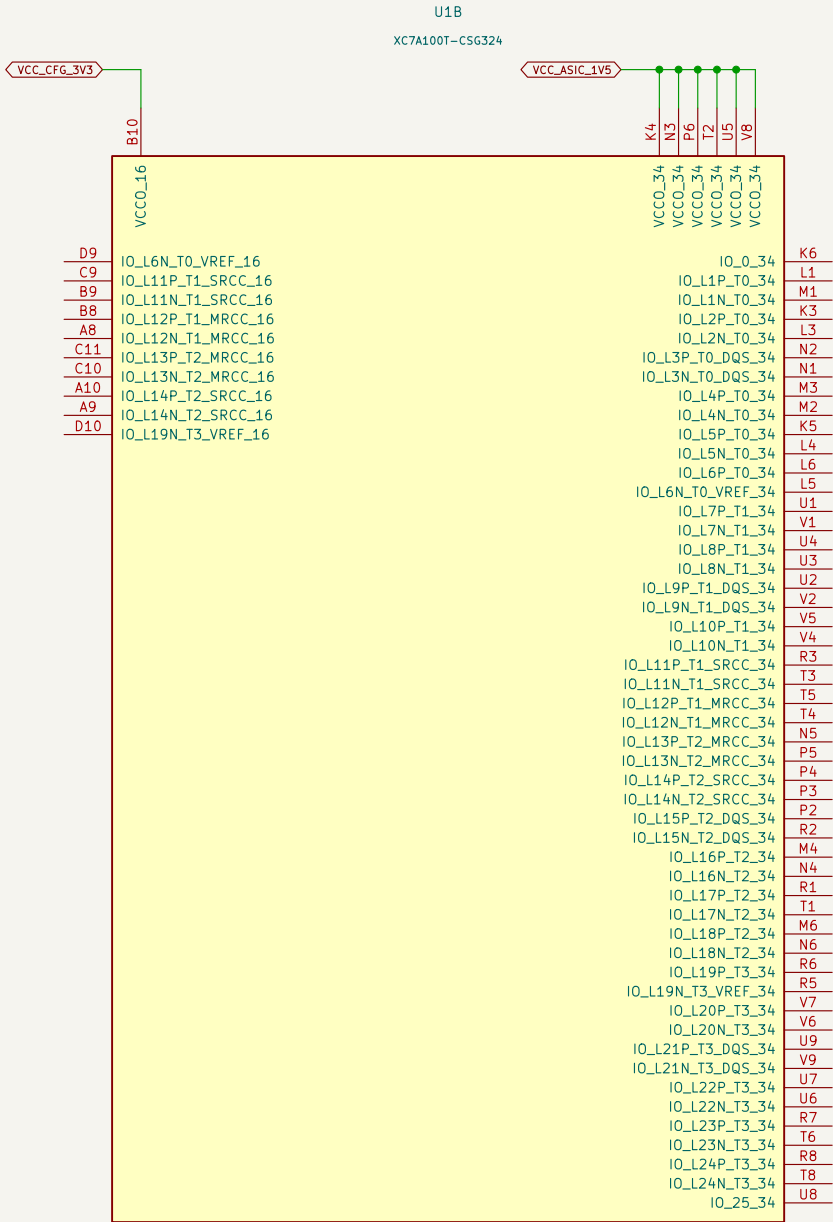
Rev: MINIMAL 0.3

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Id: 2/20

U1 / XC7A100T-CSG324 / I/O banks 16 and 34

15 x 15 mm CSG324. Unassigned I/O pins are deliberately open. The 117-signal ASIC interface is a confirmed budget; physical ASIC and receiver assignments remain deferred.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /U1 banks 16 and 34/

File: fpga_unit_02.kicad_sch

Title: U1 / XC7A100T-CSG324 / I/O banks 16 and 34

Size: A3

Date: 2026-09-26

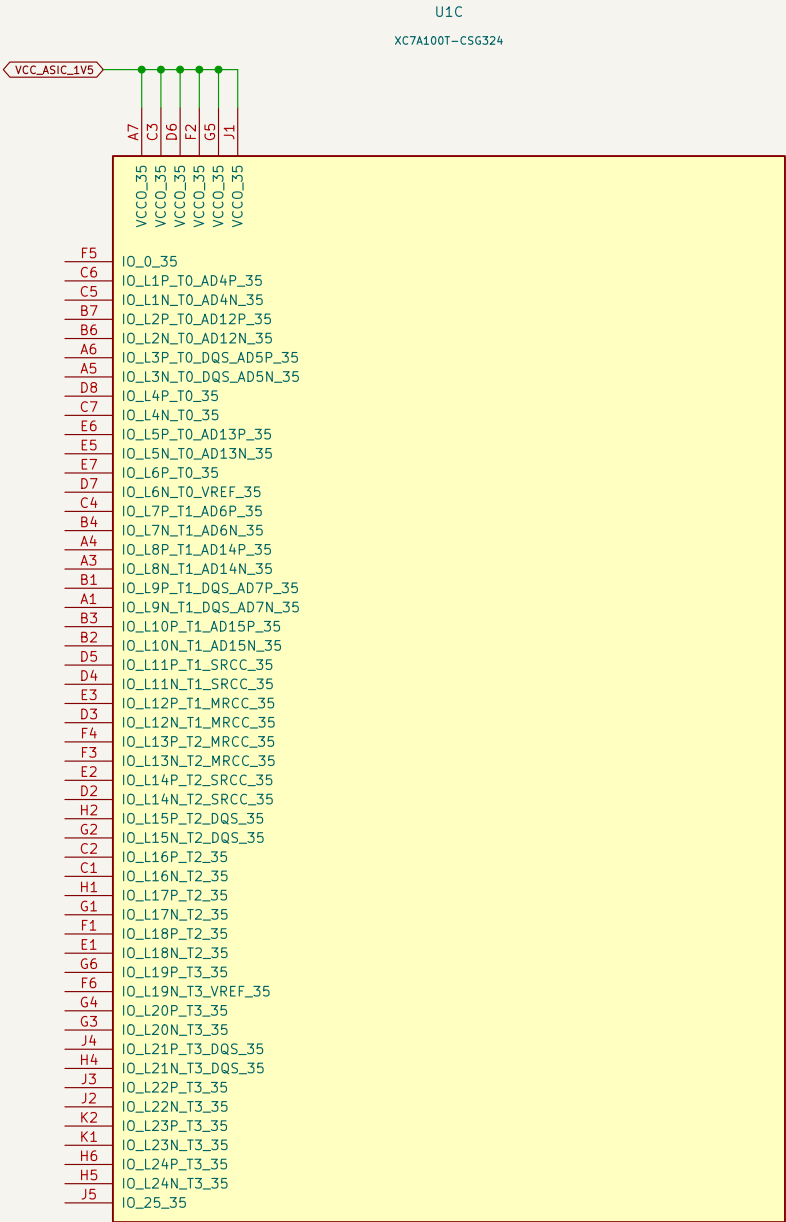
Rev: MINIMAL 0.3

KiCad E.D.A. 10.0.6

Id: 3/20

U1 / XC7A100T-CSG324 / I/O bank 35

15 x 15 mm CSG324. Unassigned I/O pins are deliberately open. The 117-signal ASIC interface is a confirmed budget; physical ASIC and receiver assignments remain deferred.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /U1 bank 35/

File: fpga_unit_03.kicad_sch

Title: U1 / XC7A100T-CSG324 / I/O bank 35

Size: A3

Date: 2026-09-26

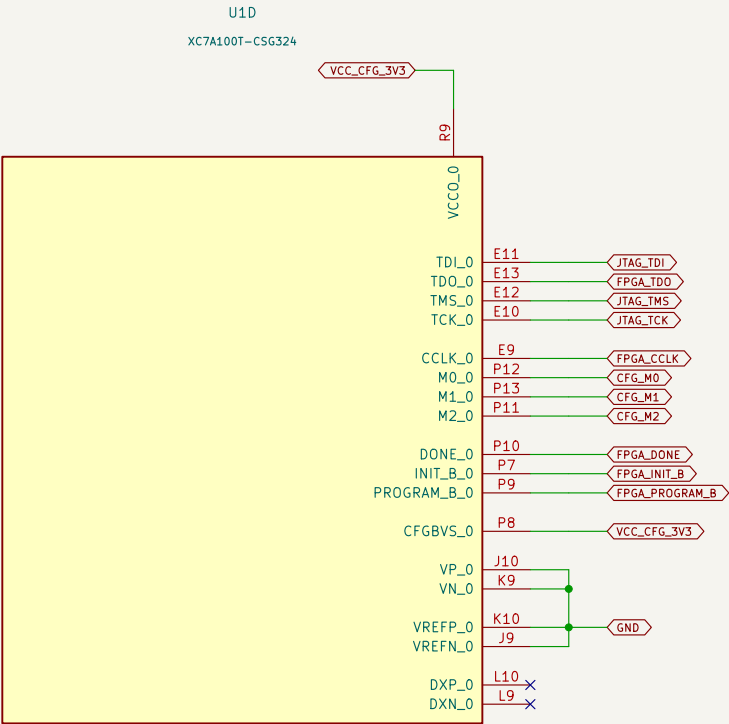
Rev: MINIMAL 0.3

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Id: 4/20

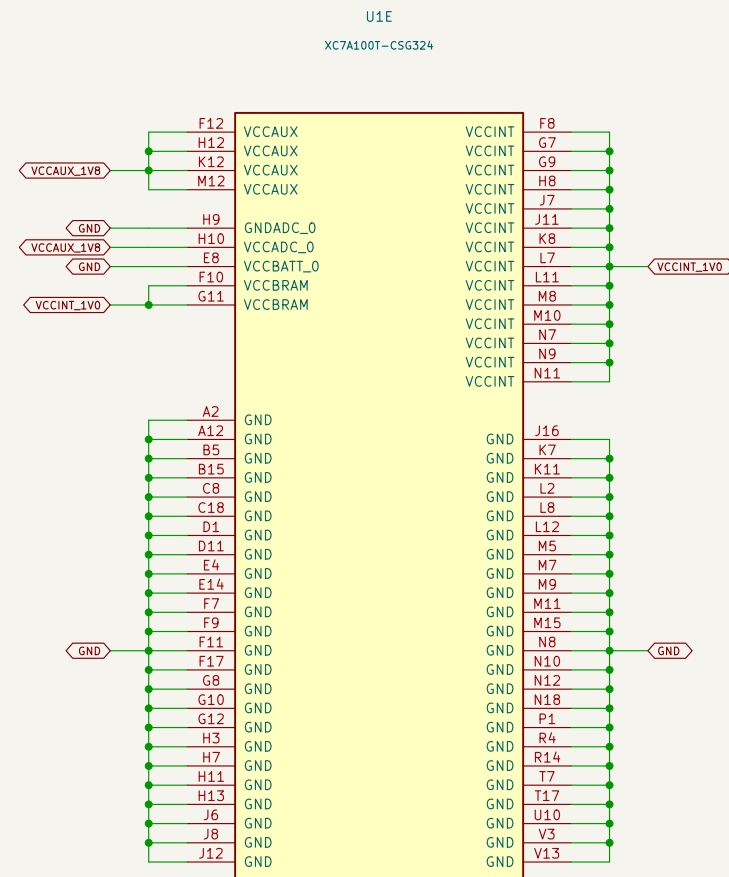
U1 / XC7A100T–CSG324 / Configuration, JTAG and analog support

15 x 15 mm CSG324. Unassigned I/O pins are deliberately open. The 117–signal ASIC interface is a confirmed budget; physical ASIC and receiver assignments remain deferred.



U1 / XC7A100T-CSG324 / Core, auxiliary power and ground

15 x 15 mm CSG324. Unassigned I/O pins are deliberately open. The 117-signal ASIC interface is a confirmed budget; physical ASIC and receiver assignments remain deferred.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /U1 core, auxiliary and ground/

File: fpga_unit_05.kicad_sch

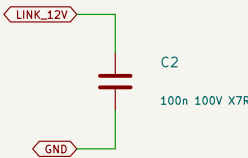
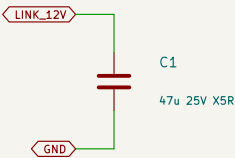
Title: U1 / XC7A100T-CSG324 / Core, auxiliary power...

Title: 01 / AC/A1001-C50524 / Core, Auxiliary power...		
Size: A3	Date: 2026-09-26	Rev: MINIMAL 03

Size: A3	Date: 2026-09-26	Rev: MINIMAL 0.3
KiCad EDA 10.0.6		Id: 6/20

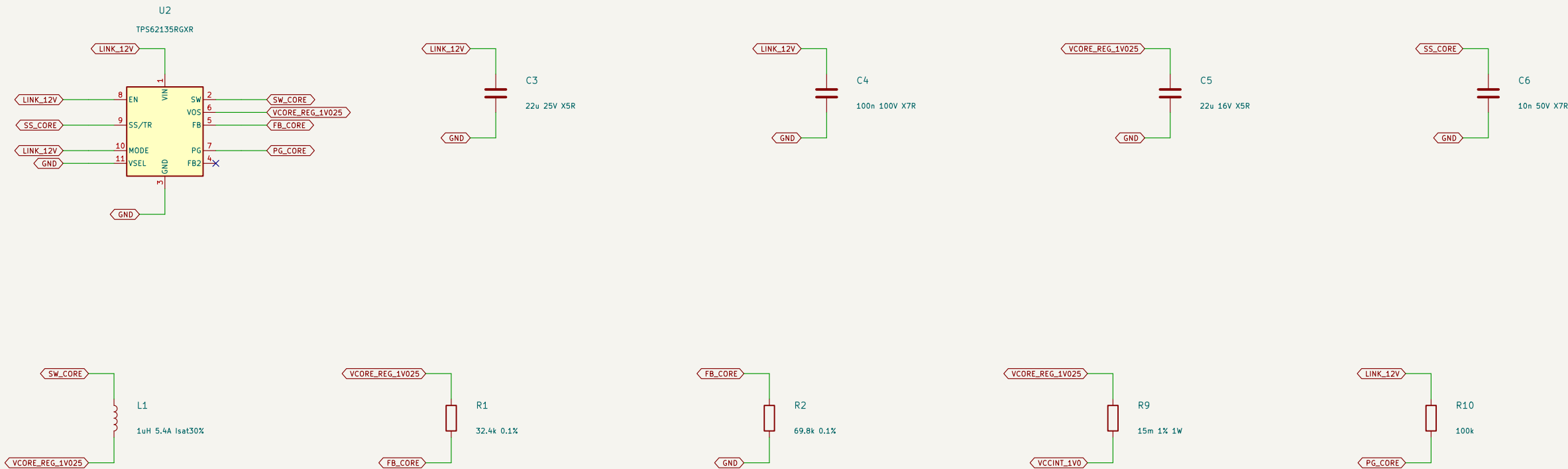
Cable input capacitors

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Core regulator / 1.0 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

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Title: Core regulator / 1.0 V

Size: A3

Date: 2026-09-26

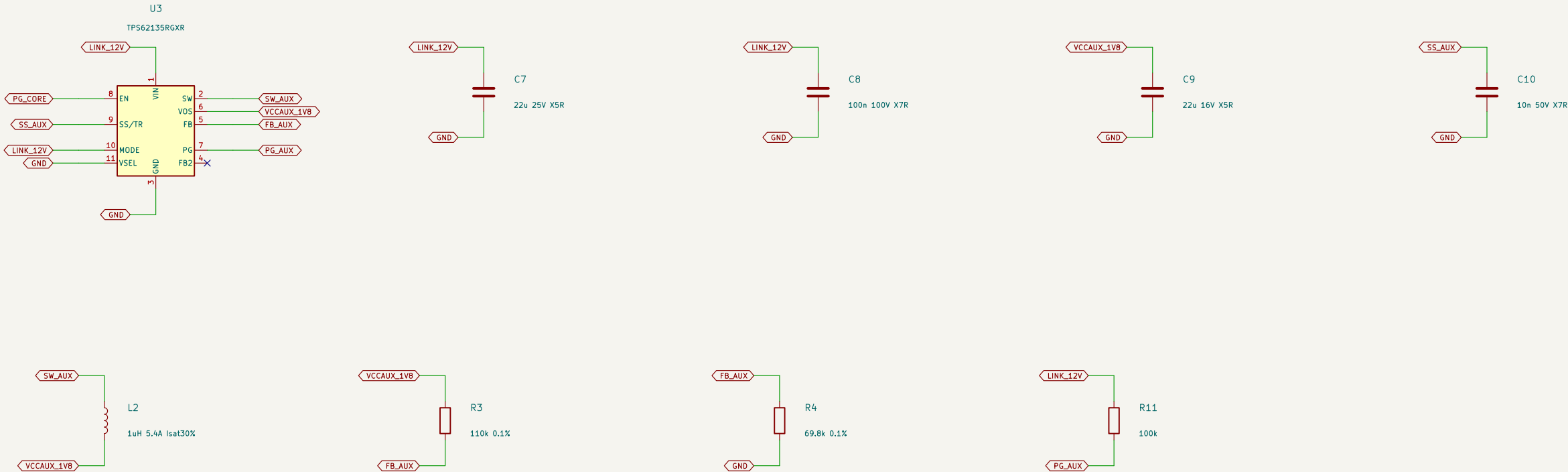
Rev: MINIMAL 0.3

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Id: 8/20

Auxiliary regulator / 1.8 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

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Size: A3

Date: 2026-09-26

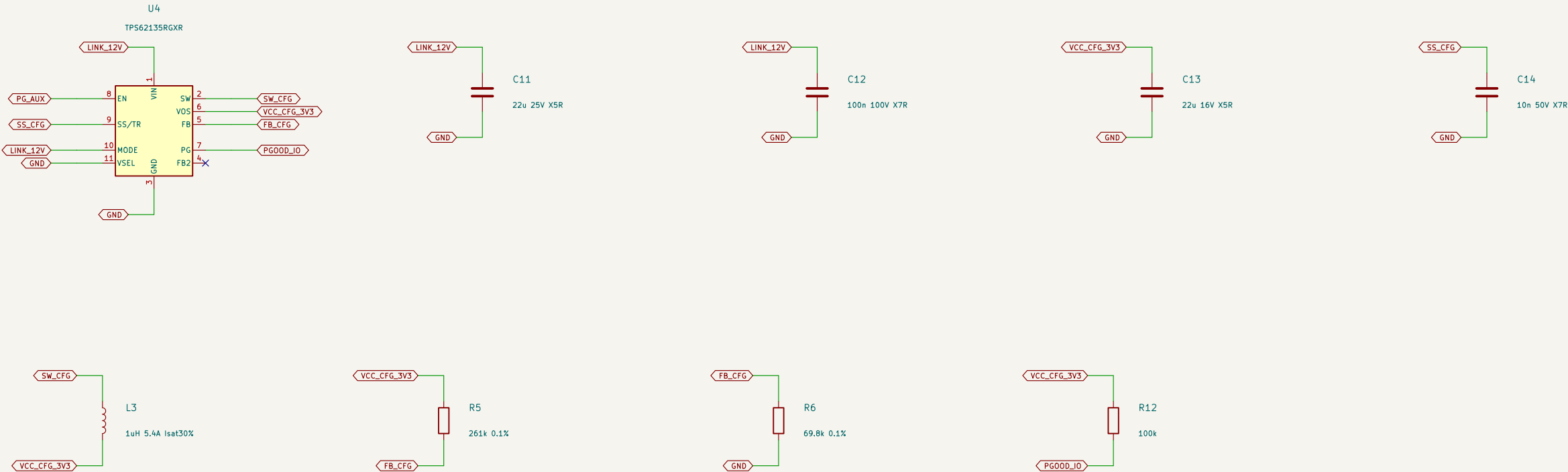
Rev: MINIMAL 0.3

KiCad E.D.A. 10.0.6

Id: 9/20

Configuration regulator / 3.3 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



ASIC bank regulator / 1.5 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.

A

B

C

D

E

F

A

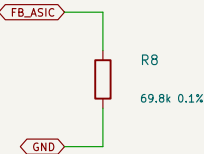
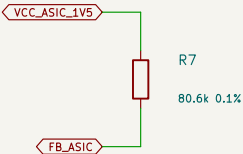
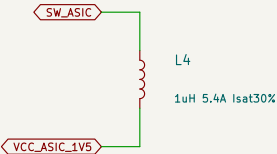
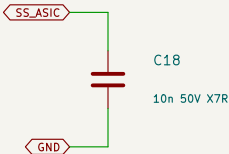
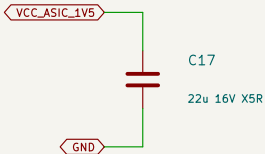
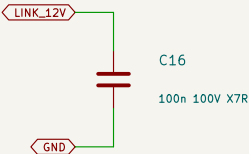
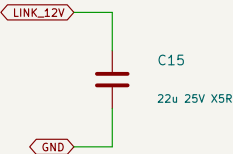
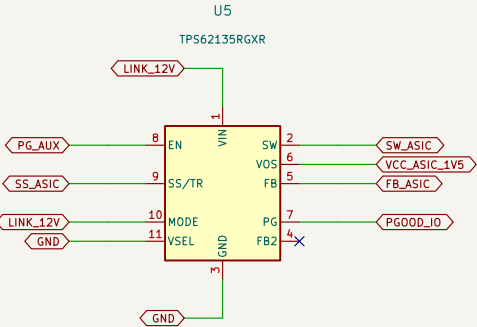
B

C

D

E

F



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

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Title: ASIC bank regulator / 1.5 V

Size: A3

Date: 2026-09-26

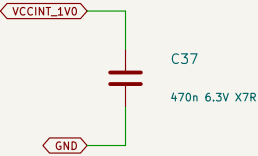
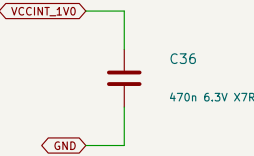
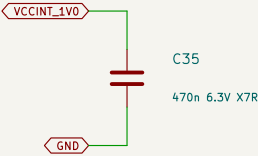
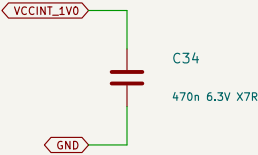
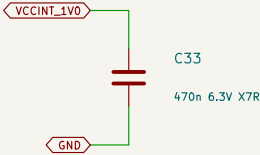
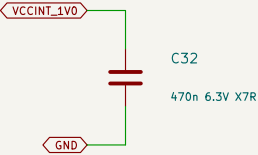
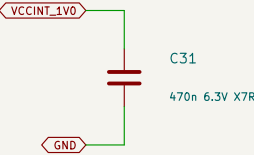
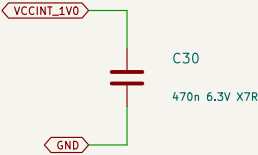
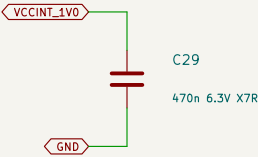
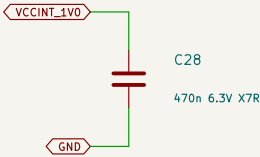
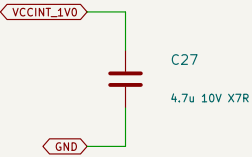
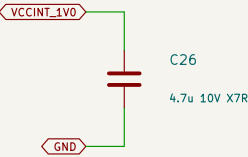
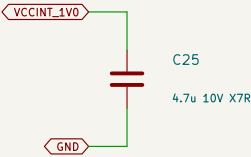
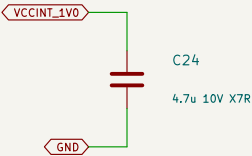
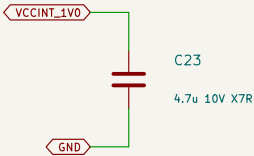
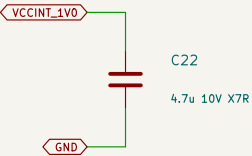
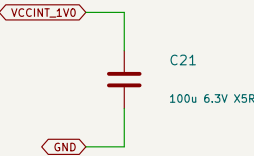
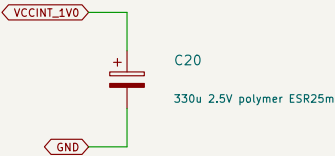
Rev: MINIMAL 0.3

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Id: 11/20

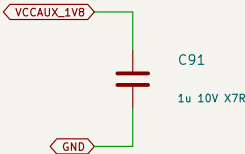
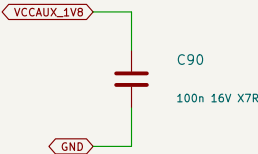
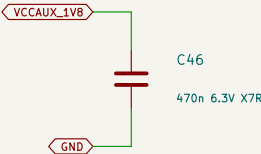
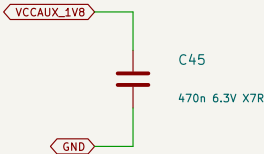
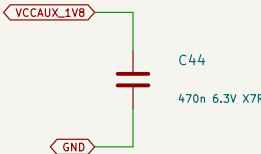
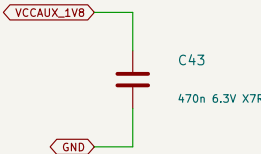
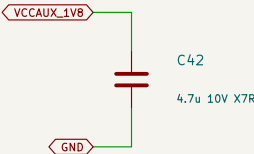
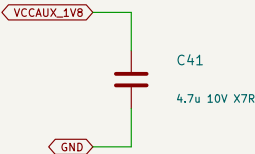
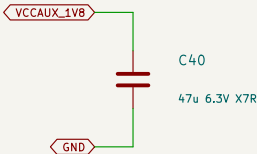
Decoupling / core and BRAM / 1.0 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



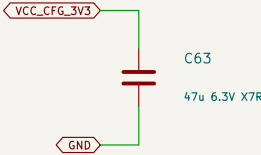
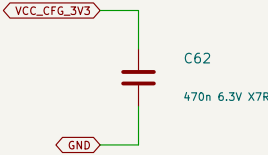
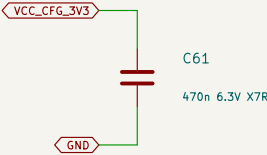
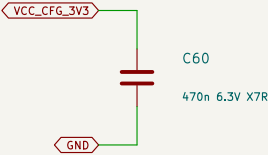
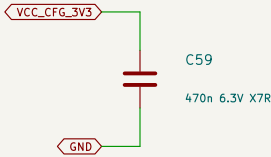
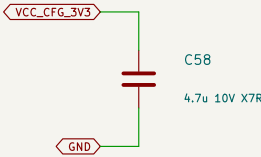
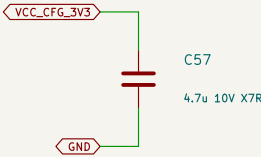
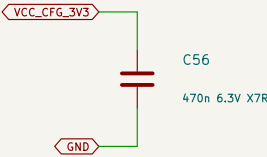
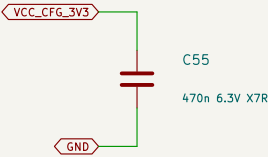
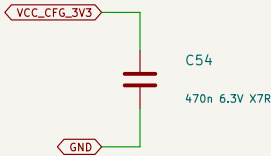
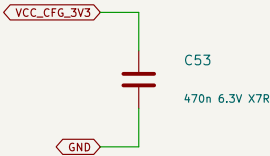
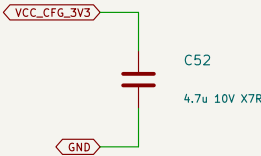
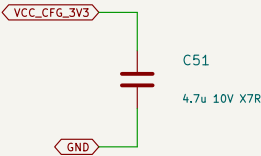
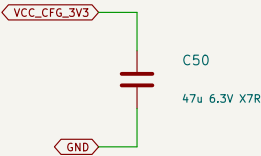
Decoupling / auxiliary and XADC / 1.8 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



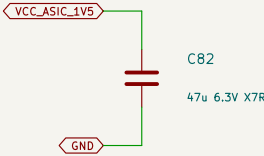
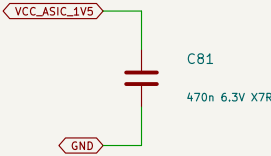
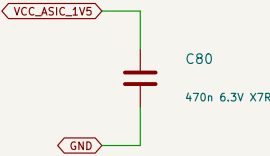
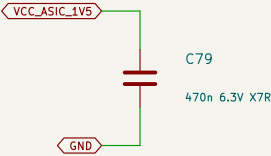
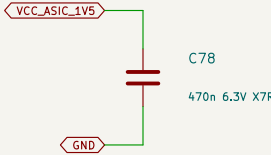
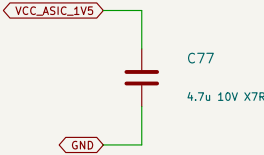
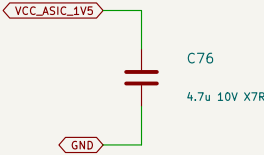
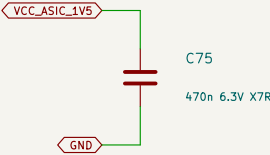
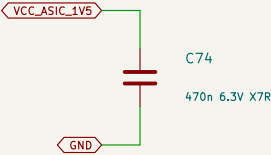
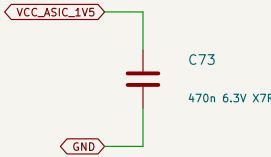
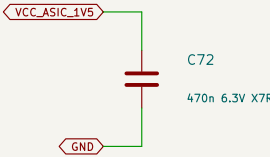
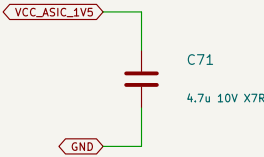
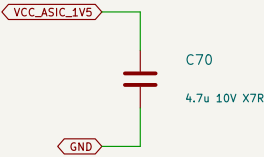
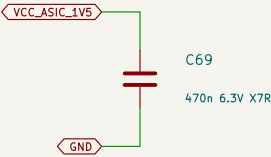
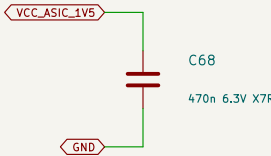
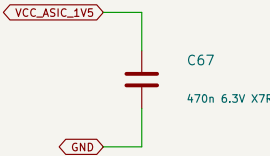
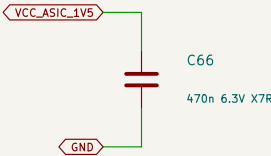
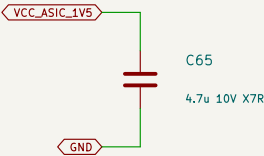
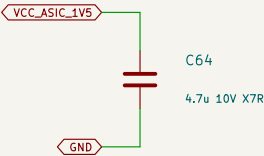
Decoupling / configuration and banks / 3.3 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



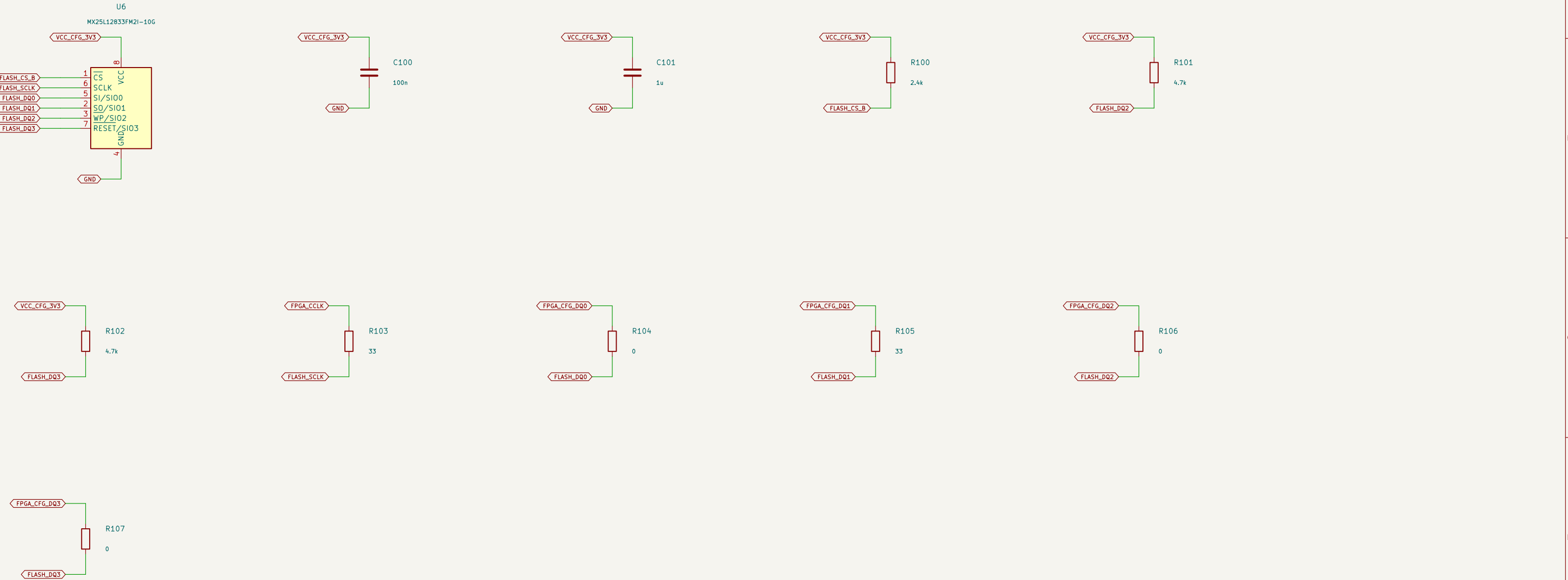
Decoupling / ASIC-facing banks / 1.5 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



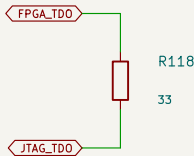
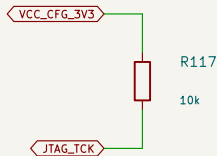
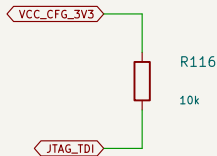
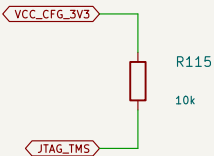
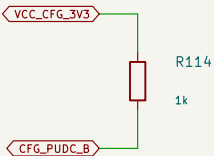
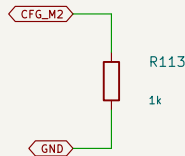
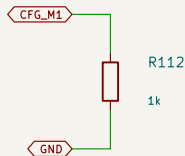
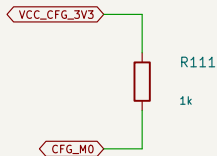
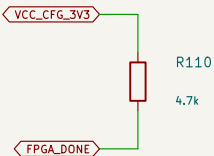
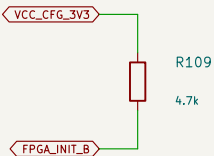
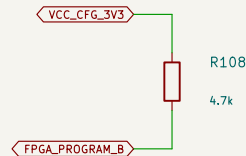
Boot flash

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Programming JTAG

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Reference clock

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.

A

B

C

D

E

F

A

B

C

D

E

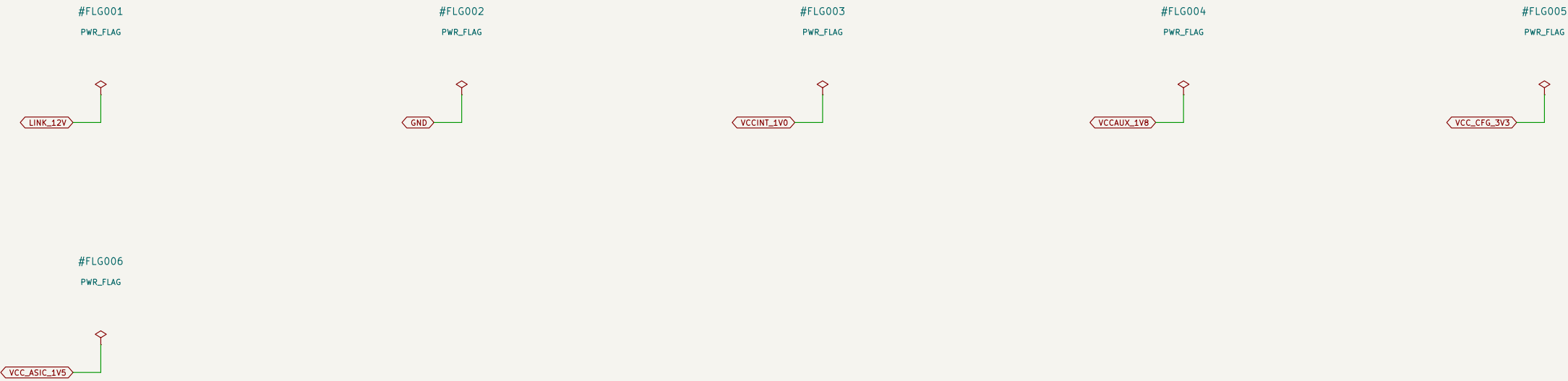
F



ENGINEERING DRAFT – NOT FOR FABRICATION		
Howard Wang / XC7A100T CSG324		
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Title: Reference clock		
Size: A3	Date: 2026–09–26	Rev: MINIMAL 0.3
KiCad E.D.A. 10.0.6	Id: 18/20	

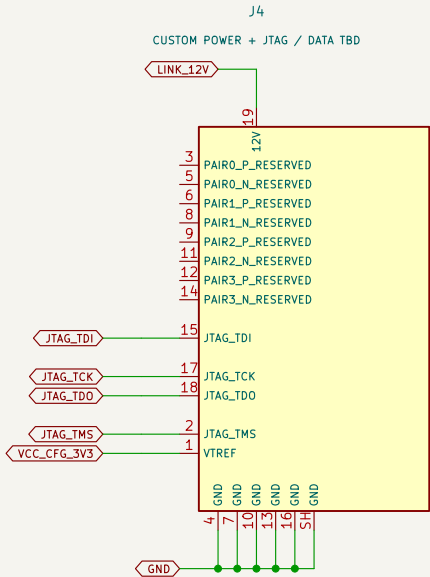
Power source assertions

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Custom shared cable / power and JTAG

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



J4 / one custom cable connector

Contact	Net	Function
19	LINK_12V	Power input, proposed nominal 12 V
16	GND	DC return
1	VCC_CFG_3V3	Target voltage reference for programmer
2	JTAG_TMS	Programmer to FPGA E12
15	JTAG_TDI	Programmer to FPGA E11
17	JTAG_TCK	Programmer to FPGA E10
18	JTAG_TDO	FPGA E13 via R118 to programmer
4, 7, 10, 13, SH	GND	Pair shields and shell

Pins 3/5, 6/8, 9/11 and 12/14: reserved data pairs, not yet assigned.

12 V is a design proposal, not a qualified cable–power specification.

Requires a protected source and verified cable/contact current ratings.

Do not connect this custom–powered port to ordinary HDMI equipment.

No separate JTAG header, boot–mode jumper or test points.